Assignment stage 2 description

Start Assignment

- Due No due date
- Points 0
- Submitting a text entry box

Stage 2 (30 marks, due Week 12)

The objective of Stage 2 is to complete the analysis and model building to solve the problem defined in Stage 1, demonstrating both collaborative design and individual technical contribution.

Core Idea

- Each student works on their own dataset (from Stage 1)
- Each student applies the same three models
- The group compares results across all datasets
- Final outcome: identify the best model overall

Project Structure

- **Group component:** shared problem, framework, and evaluation design
- **Individual component:** each student implements and evaluates their own models on their dataset

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Components

This stage consists of:

1. Group Component (14 marks, ~1000 words ±10%)

- Literature Review – 6 marks
- Experimental Setup – 4 marks
- Cross-Dataset Comparison & Final Recommendation – 4 marks

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2. Individual Component (16 marks, ~800 words ±10% per student)

- Applying 3 Models – 9 marks
- Results & Interpretation – 7 marks

Report Structure (Combined Submission)

Length

- ~3400-4200 words (±10%)
- One group report with clearly labelled individual sections
- Appendix, visualisations, and references excluded from word count

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Section 1: Literature Review (Group) – 6 marks

- 6–8 papers covering the three selected models
- Explain why these models were chosen

Section 2: Experimental Setup (Group) – 4 marks

- Research question(s)
- Hypotheses
- Description of the three models
- Evaluation metrics (e.g., F1, RMSE, R^2)
- Validation method (e.g., cross-validation)

Section 3: Implementation (Individual) – 9 marks

Each student must include:

Student Name: [Name]

- Dataset description (brief)
- Application of all three models
- Preprocessing & feature engineering
- Parameter tuning (if applicable)

Section 4: Results & Comparison (Individual) – 7 marks

Each student must include:

Student Name: [Name]

- Results for all three models (tables/figures)
- Performance comparison
- Best model for their dataset
- Brief justification

Section 5: Group Comparison & Conclusion (Group) – 4 marks

As a group:

- Compare results across all datasets
- Identify:
  - Best performing model overall
  - Whether performance depends on dataset characteristics
- Final recommendation:
  - Best model
  - When to use each model

Criterion	Marks	Excellent	Good	Satisfactory	Poor / Not Met
Literature Review (Group)	6	6–8 highly relevant papers clearly summarised; strong justification for chosen models	Relevant papers with reasonable explanation	Papers included but weak relevance or limited explanation	Minimal or no literature review
Experimental Setup (Group)	4	Clear research questions, hypotheses, models, metrics, and validation; well connected	Most elements clear with minor gaps	Some elements missing or unclear	Poorly defined or missing setup
Cross-Dataset Comparison & Recommendation (Group)	4	Strong comparison across datasets; clear best model; insightful recommendations	Good comparison with reasonable conclusions	Limited comparison; weak conclusions	Minimal or no comparison
Applying 3 Models (Individual)	9	All 3 models correctly implemented; strong preprocessing, tuning, and understanding	All models applied with minor issues	Models applied with errors or limited effort	Incomplete or incorrect implementation
Results & Interpretation (Individual)	7	Clear results (tables/figures); strong	Clear results with reasonable interpretation	Basic results; limited analysis	Poor or missing results/analysis

Criterion	Marks	Excellent	Good	Satisfactory	Poor / Not Met
		comparison; insightful interpretation			
Stage 2 Rubric (group & ind - 30 Marks)					

Criteria	Ratings				Pts
Literature Review (Group) – 6 marks threshold: 6.0 pts	**6 Pts** **Excellent: 6–8 highly relevant papers clearly summarised; strong justification for chosen models**	**4 Pts** **Good: Relevant papers included with reasonable explanation**	**2 Pts** **Satisfactory: Papers included but weak relevance or limited explanation**	**1 Pts** **Poor / Not Met: Minimal or no literature review**	6 pts
Experimental Setup (Group) – 4 marks threshold: 4.0 pts	**4 Pts** **Excellent: Clear research questions, hypotheses, models, metrics, and validation; all well connected**	**3 Pts** **Good: Most elements clear with minor gaps**	**2 Pts** **Satisfactory: Some elements missing or unclear**	**1 Pts** **Poor / Not Met: Poorly defined or missing setup**	4 pts
Cross-Dataset Comparison & Recommendation (Group) – 4 marks threshold: 4.0 pts	**4 Pts** **Excellent: Strong comparison across datasets; clear best model; insightful recommendations**	**3 Pts** **Good: Good comparison with reasonable conclusions**	**2 Pts** **Satisfactory: Limited comparison; weak conclusions**	**1 Pts** **Poor / Not Met: Minimal or no comparison**	4 pts
Applying 3 Models (Individual) – 9 marks threshold: 9.0 pts	**9 Pts** **Excellent: All 3 models correctly implemented; strong preprocessing, tuning, and understanding**	**7 Pts** **Good: All models applied with minor issues**	**5 Pts** **Satisfactory: Models applied with errors or limited effort**	**3 Pts** **Poor / Not Met: Incomplete or incorrect implementation**	9 pts
Results & Interpretation (Individual) – 7 marks threshold: 7.0 pts	**7 Pts** **Excellent: Clear results (tables/figures); strong comparison; insightful interpretation**	**5 Pts** **Good: Clear results with reasonable interpretation**	**3 Pts** **Satisfactory: Basic results; limited analysis**	**1 Pts** **Poor / Not Met: Poor or missing results/analysis**	7 pts
					Total points: 30